

CE3420

Assignment

Name: Kaushal Singh

Roll Number: CH22BTECH11018

Working with Geopandas

```
In [1]: """ Question
You will notice that the output file contains a row with the home_city as
Modify the filtered dataframe to remove this row and write the results to
Hint: Use the Boolean filtering method we learnt earlier to select rows t
"""

import os
from geopy import distance
import pandas as pd

def calculate_distance(row):
    city_coordinates = (row['lat'], row['lng'])
    return distance.geodesic(city_coordinates, home_city_coordinates).km

data_pkg_path = 'python_foundation/data'
filename = 'worldcities.csv'
path = os.path.join(data_pkg_path, filename)
df = pd.read_csv(path)

home_country = 'India'
home_city = 'Bengaluru'

country_df = df[df['country'] == home_country].copy()
filtered = country_df[country_df['city_ascii'] == home_city]
home_city_coordinates = (filtered.iloc[0]['lat'], filtered.iloc[0]['lng'])

country_df['distance'] = country_df.apply(calculate_distance, axis=1)
filtered = country_df[['city_ascii', 'distance']]
filtered = filtered.rename(columns={'city_ascii': 'city'})
filtered = filtered[filtered['city'] != home_city]

output_path = os.path.join('Assignment3', 'distances.csv')
filtered.to_csv(output_path, index=False)
```

```
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/urllib3/
__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1
+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: http
s://github.com/urllib3/urllib3/issues/3020
warnings.warn(
```

Creating Spatial Data

```
In [2]: """ Question
The data package contains multiple geonames text files from different cou
We have the code below that reads all the files, extract the mountain fea
```

The exercise is to convert the merged DataFrame to a GeoDataFrame as save
"""

```
import os
import pandas as pd

# %pip install geopandas
import geopandas as gpd

data_pkg_path = 'python_foundation/data/geonames/'
files = os.listdir(data_pkg_path)

column_names = [
    'geonameid', 'name', 'asciiname', 'alternatenames',
    'latitude', 'longitude', 'feature class', 'feature code',
    'country code', 'cc2', 'admin1 code', 'admin2 code',
    'admin3 code', 'admin4 code', 'population', 'elevation',
    'dem', 'timezone', 'modification date'
]

dataframes = []
for file in files:
    path = os.path.join(data_pkg_path, file)
    df = pd.read_csv(path, sep='\t', names=column_names)
    mountains = df[df['feature class'] == 'T']
    dataframes.append(mountains)

merged = pd.concat(dataframes)

geometry = gpd.points_from_xy(merged['longitude'], merged['latitude'])
gdf = gpd.GeoDataFrame(merged, geometry=geometry, crs='EPSG:4326')

output_path = os.path.join('Assignment3', 'mountains.shp')
gdf.to_file(output_path)
```

```

/var/folders/2c/6txt2yzx14s32xy8mxjtdvw0000gn/T/ipykernel_20439/261063226
1.py:28: DtypeWarning: Columns (9,11) have mixed types. Specify dtype opti
on on import or set low_memory=False.
    df = pd.read_csv(path, sep='\t', names=column_names)
/var/folders/2c/6txt2yzx14s32xy8mxjtdvw0000gn/T/ipykernel_20439/261063226
1.py:28: DtypeWarning: Columns (9) have mixed types. Specify dtype option
on import or set low_memory=False.
    df = pd.read_csv(path, sep='\t', names=column_names)
/var/folders/2c/6txt2yzx14s32xy8mxjtdvw0000gn/T/ipykernel_20439/261063226
1.py:28: DtypeWarning: Columns (9,13) have mixed types. Specify dtype opti
on on import or set low_memory=False.
    df = pd.read_csv(path, sep='\t', names=column_names)
/var/folders/2c/6txt2yzx14s32xy8mxjtdvw0000gn/T/ipykernel_20439/261063226
1.py:38: UserWarning: Column names longer than 10 characters will be trunc
ated when saved to ESRI Shapefile.
    gdf.to_file(output_path)
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'alternatenam
es' to 'alternaten'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'feature clas
s' to 'feature cl'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'feature cod
e' to 'feature co'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'country cod
e' to 'country co'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'admin1 code'
to 'admin1 cod'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'admin2 code'
to 'admin2 cod'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'admin3 code'
to 'admin3 cod'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'admin4 code'
to 'admin4 cod'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Normalized/laundered field name: 'modification
date' to 'modificati'
    ogr_write(
/Users/17kaushalsingh/Library/Python/3.9/lib/python/site-packages/pyogrio/
raw.py:723: RuntimeWarning: Value 'Bliihi Islands,Bliinies Islands,Blijni
Islands,Blijnies Islands,Blizhnie Islands,Blizhnie ostrova,Blizhniye Islan
ds,Iilhas Near,Islas Near,Isole Near,Merjaka kgziner,Near Adalari,Near Adal
ari,Near Islands,Nearsaaret,Plishnie Ostrova,Plishnie Ostrowa Islands,Qaxu
m Tanangis,Rat Islands,Sasignan Islands,Sasignan Tanangin,Sasignan tanangi
n,Sasignanoj,Wyspy Bliskie,ni er qun dao,nia zhu dao,Ближние острова,Մերձ
ակա կղզիներ,მახლობელი კუნძულები,ニア , ' of field alternaten has

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been truncated to 254 characters. This warning will not be emitted any more for that layer.
ogr_write(

Introduction to NumPy

```
In [3]: """ Question
Sum the array b along Axis=1. What do you think will be the result?

import numpy as np

b = np.array([[1, 2, 4], [3, 4, 5]])
b
"""

import numpy as np

b = np.array([[1, 2, 4], [3, 4, 5]])
result = b.sum(axis=1)
result
```

Out[3]: array([7, 12])

Working with RasterIO

```
In [4]: """ Question
The merged array represents elevation values. The extent of the tiles cov

import rasterio
import os
import numpy as np

output_filename = 'merged.tif'
output_dir = 'output'
output_path = os.path.join(output_dir, output_filename)

# Read the output file as a NumPy array and find the maximum value"""

# %pip install rasterio
import rasterio
import os
import numpy as np

output_filename = 'merged.tif'
output_dir = 'python_foundation/output'
output_path = os.path.join(output_dir, output_filename)

with rasterio.open(output_path) as src:
    data = src.read(1)
    nodata = src.nodata

data = data[data != nodata]
max_value = np.max(data)
max_value
```

Out[4]: np.int16(8748)

Writing Standalone Python Scripts

```
In [5]: """  
        No exercise in this section  
        """
```

```
Out[5]: '\nNo exercise in this section\n'
```

```
In [ ]:
```